MACQUARIE University

Progression and Retention Analytics

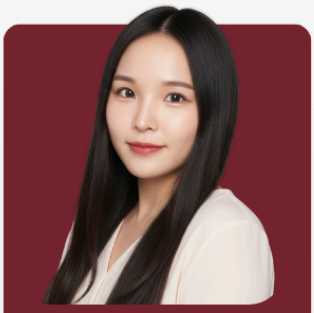
Informing Course Design and Intervention Strategies for Macquarie Business School Course Directors

Prepared by

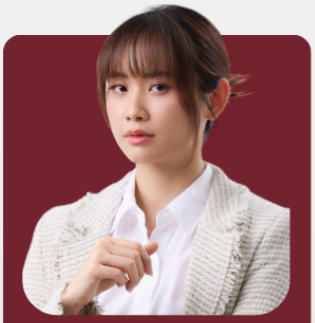
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Executive Summary

This report provides Course Directors with decision-ready evidence to reduce progression delays and attrition in MQBS programmes. The analysis examines the 2020–2025 student dataset, develops two dashboards, and trains a logistic regression model to identify high-risk cohorts for proactive intervention.

Core insights

1. **First-year bottlenecks:** Foundational units create the primary structural barrier and generate the highest fail counts, creating pipeline choke points that delay completion.
2. **Performance signal:** Students who drop out typically record multiple repeat fails and fall within the low to mid WAM band. Completers typically have few or no repeats and achieve higher WAMs.
3. **Engagement risk:** Very low study loads in part-time students correlate with attrition. Risk drops once the load exceeds 0.5 EFTSL per year.
4. **Predictive targeting:** The model achieved approximately 86% accuracy (AUC 0.90) and identified 285 students (approximately 4% of 7,079 active in 2025) with over 80% dropout probability for early outreach.

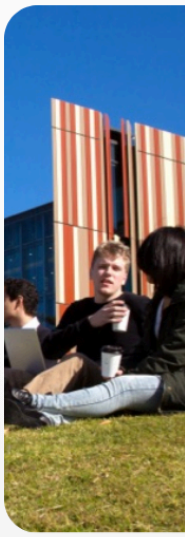
Recommendations

1. **Review high-risk foundation units and redesign accordingly** through granular analysis of individual units and tailored interventions.
2. **Enhance first-year transition support** by establishing peer mentoring programs and expanding O-Week programming on university navigation and support service access.
3. **Implement tiered intervention protocols** by deploying the predictive model to stratify students into three risk categories with differentiated support strategies for each tier.

These measures shift practice from reactive remediation to proactive, data-driven course management. This approach improves on-time completion and regulatory alignment while strengthening student outcomes.



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Introduction

Background and Context

Although Macquarie University reported a success rate of 88.12% in 2024, slightly below the NSW state average of 89.07% (Department of Education, 2025), the university must continue to monitor student progression and recognise opportunities to improve student retention and success rates. These are not merely statistics but critical factors affecting institutional performance, government funding allocations, and long-term student outcomes

The university's Retention Taskforce, launched in 2023, addresses these challenges by identifying at-risk students early, providing timely feedback, and implementing targeted support interventions, aligning with the Higher Education Standards Framework Standards 1.3.3 to 1.3.6 and Standard 8 of the National Code 2018 (Department of Education, 2024; TEQSA, 2022).

This research supports the initiative by identifying evidence-based strategies to improve retention and success rates, which will benefit students through enhanced learning experiences, faculty through effective resource allocation, and the institution through stronger performance metrics and regulatory compliance.

Research Scope and Focus

Course Directors within the Macquarie Business School lack visibility into early warning signals that predict student attrition and progression delays. Without clear insights into which unit combinations, progression pathways, and student study behaviours differentiate successful students from those who withdraw or take longer to complete, Course Directors struggle to make informed decisions about curriculum design, sequencing, and targeted intervention strategies.

This project addresses these challenges by examining historical enrolment and retention data to uncover patterns that facilitate or hinder student progression and retention. The analysis scope includes undergraduate students enrolled in Macquarie Business School courses between 2020 and 2025.

The project delivers three key outputs tailored to the Course Director persona. First, two interactive Power BI dashboards provide insights into student progression pathways and study patterns, as well as retention and attrition analysis. Second, a predictive analytics model using logistic regression forecasts which students from the current cohort (those continuing and commencing in 2025) are likely to be retained through to 2026, enabling proactive intervention strategies. These outputs address two critical problem statements, broken down into specific analytical questions as shown in Table 1.

Table 1

Problem Statements and Research Questions

Problem Statements	1. Course Directors struggle to identify specific high-failure units and problematic unit sequences that significantly prolong student study duration or prevent timely course completion, hindering curriculum optimisation and the provision of structural support.	2. Course Directors lack visibility into early warning signals that indicate students are at risk of dropping out, limiting their ability to implement timely interventions.
Research Questions	• Which unit has the highest failure rate (severity)? Which unit has the highest absolute fail count (impact)? What level is this unit? • Which specific units are the most persistent traps (highest count of repeat failures), and what is the average progress percentage of students stuck in them? • What are the factors taken by students who complete on-time vs. those who take longer? • At what point in the course journey do most students experience delays? • Are certain student groups disproportionately failing the high-risk units?	• To what extent do patterns of study intensity serve as early behavioural indicators of attrition? • How do variations in academic performance interact to differentiate students who complete from those who dropped out? • What demographic profiles tend to correspond with higher likelihoods of dropping out?

Data and Methodology

Data Sources and Description

The dataset used for this project consists of de-identified student enrolment and retention data provided by the Macquarie University Business Intelligence and Reporting. The data covers the period 2020-2025 and includes undergraduate students enrolled in core and elective units at the Business School. Key fields include student identifiers, academic year, unit codes, grades, attendance mode, course information, demographic data (gender, age, citizenship), credit points, weighted average marks, and retention status indicators. The dataset contains 284,656 records representing individual unit enrollments from 21,814 unique students across multiple courses.

Data Cleaning and Preparation

Missing values were systematically addressed. GRADE missing data for withdrawn units were imputed as 'WD', TRANSFER-related fields were filled with 'No Transfer', and missing COURSE_COMPLETION_DATE values were transformed into a binary COURSE_COMPLETED feature to indicate completion status directly. Data validation included checking for inconsistent data types across 54 columns, verifying data logic, and ensuring temporal consistency. Standardisation was applied to categorical variables through label encoding, while numerical features like credit points and marks were scaled

appropriately. Exclusion criteria focused on removing records with critical missing values in core fields, such as student key identifiers, that couldn't be reasonably imputed, as these would hinder accurate student-level aggregation.

Feature Engineering

Multiple new variables were created to enhance predictive capability. Progress and Performance Metrics included PROGRESS_PERCENTAGE (completed vs total credit points) to standardize academic standing. Temporal and Workload Features comprised STUDY_DURATION_YEARS (for completed courses), and YEAR_EFTSL_LOAD (academic workload measure). Contextual Categorisations included UNIT_LEVEL_CATEGORY and demographic attributes. Student-level Aggregates captured failure patterns, withdrawal history, and course load variability. The comprehensive AT_RISK indicator combined multiple risk factors into a single predictive metric. These engineered features enable the model to identify subtle patterns in student progression that raw data alone cannot reveal, significantly enhancing predictive capability for early intervention.

Key Summary Statistics

The dataset comprises 284,656 unit enrolment records from 21,814 unique students across 85 distinct courses. Almost half of the cohort were identified as



academically at risk, primarily due to repeated course failures and low weighted average marks (WAMs). Overall academic progress varied widely across programs, reflecting diverse study patterns and levels of engagement. While some students demonstrated consistent achievement and timely completion, others showed signs of slowed progression and reduced academic momentum.

These findings highlight the complex interplay between academic performance, study behaviour, and demographic factors in shaping student success. They provide a foundation for the subsequent dashboards, which examine how curriculum structure and behavioural patterns collectively influence attrition and progression outcomes.

Data Analysis and Insights

Overview

This analysis explores factors influencing student progression and retention, and attrition behavior within Macquarie Business School's undergraduate programs. Our approach comprises two interactive dashboards for visual data exploration and a logistic regression model for predictive insights (Complete files in Appendix A). The following section presents both dashboard designs and functionality, followed by the logistic regression methodology and rationale.

Dashboard 1: Curriculum Barrier Analysis

Research Question and Purpose

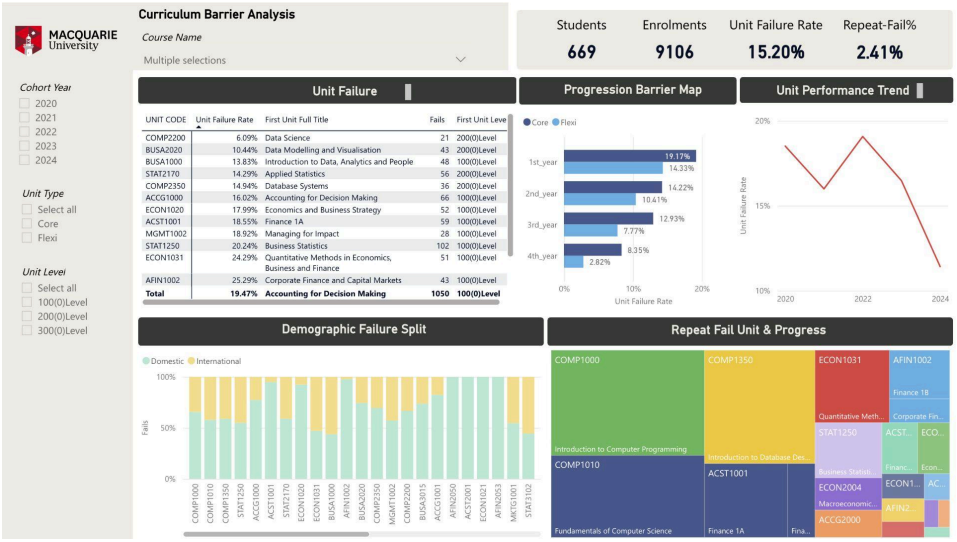
The dashboard targets problem statement 1, addressing structural barriers in the curriculum that hinder timely course completion. It provides Course Directors with systematic tools to evaluate course design, identifies peak academic pressure points by mapping failure concentrations, and quantifies the costs of delays by highlighting units with high repeat failure rates.

Dashboard Design and Layout

As shown in Figure 1, the dashboard guides Course Directors through a structured analytical process by first selecting cohort year and course using the top filters. The layout begins with Key Performance Indicator cards providing an immediate overview of student numbers, enrolments, and failure rates. The Unit Failure table identifies units presenting the greatest barriers ranked by failure rate, while the Progression Barrier Map shows how failure rates vary across year levels and the Unit Performance Trend tracks changes over time. The Demographic Failure Split reveals patterns between domestic and international students, and the Repeat Fail analysis highlights units where students struggle repeatedly, enabling Course Directors to identify problems and target interventions effectively.

Figure 1

Dashboard 1 Overview: Curriculum Barrier Analysis



Key Findings and Interpretations

Unit Failure Rate. The Unit Failure table (Figure 2) identifies where interventions are most needed. Analysis of units from 2020 to 2024 reveals that the highest failure rates and highest total number of failures occur predominantly at the foundational level. The top three are COMP1010, COMP1000, and COMP1350, with the first two representing the largest bottlenecks in the programme. This pattern demonstrates that the most urgent structural issue is concentrated at entry level. Addressing these foundational barriers should be the primary focus for intervention efforts.

Figure 2

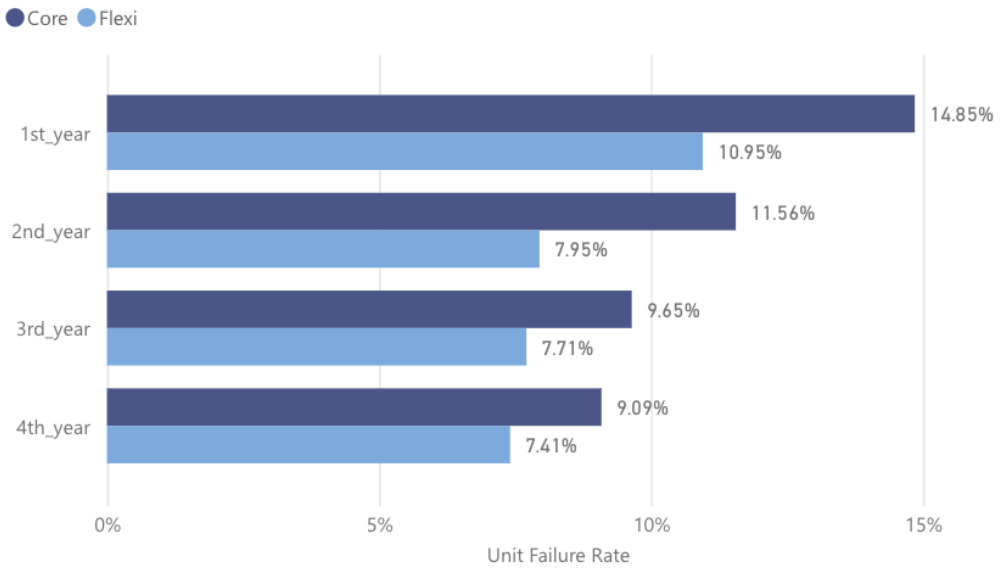
Dashboard 1: Unit Failure Rate

UNIT CODE	Unit Failure Rate	First Unit Full Title	Fails	First Unit Level
COMP1010	30.00%	Fundamentals of Computer Science	147	100(0)Level
COMP1000	26.91%	Introduction to Computer Programming	169	100(0)Level
COMP1350	26.43%	Introduction to Database Design and Management	129	100(0)Level
AFIN1002	25.29%	Corporate Finance and Capital Markets	43	100(0)Level
ECON1031	24.29%	Quantitative Methods in Economics, Business and Finance	51	100(0)Level
STAT1250	20.24%	Business Statistics	102	100(0)Level
MGMT1002	18.92%	Managing for Impact	28	100(0)Level
ACST1001	18.55%	Finance 1A	59	100(0)Level
ECON1020	17.99%	Economics and Business Strategy	52	100(0)Level
ACCG1000	16.02%	Accounting for Decision Making	66	100(0)Level
BUSA1000	13.83%	Introduction to Data, Analytics and People	48	100(0)Level

Progression Barrier Map. The Progression Barrier Map (Figure 3) reinforces the findings from Figure 2. The data reveals that academic pressure is highest upon entry, with first-year core units recording the highest failure rate. This rate declines substantially in second year and continues improving in third and fourth year. This pattern confirms that the curriculum's main structural barrier occurs in the first year, when students face the most significant hurdle. The findings suggest a misalignment between the assumed knowledge for 100-level units and actual student capability, indicating that prerequisites and entry-level curriculum design require structural reform to better support incoming students.

Figure 3

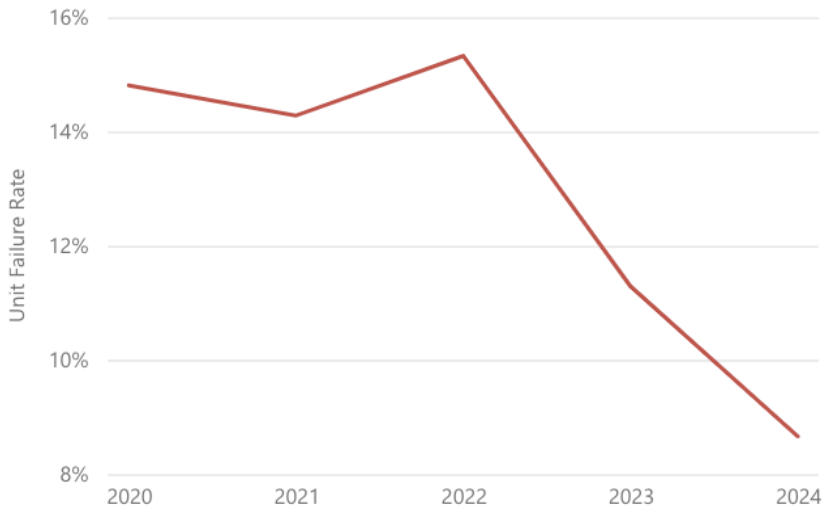
Dashboard 1: Progression Barrier Map



Unit Performance Trend. As illustrated in Figure 4, the Unit Performance Trend for the Business Analytics course exemplifies how failure rates have fluctuated over time. The programme-wide failure rate was high in 2020 and spiked again in 2022. This suggests the programme was susceptible to external factors such as pandemic effects. However, since 2022 the trend has been positive, with significant improvement in 2024. This suggests that interventions or curriculum changes implemented after the 2022 spike may already be taking effect, leading to more successful outcomes for recent cohorts. This pattern demonstrates the importance of monitoring unit performance over time to evaluate the effectiveness of structural reforms and identify periods requiring additional support.

Figure 4

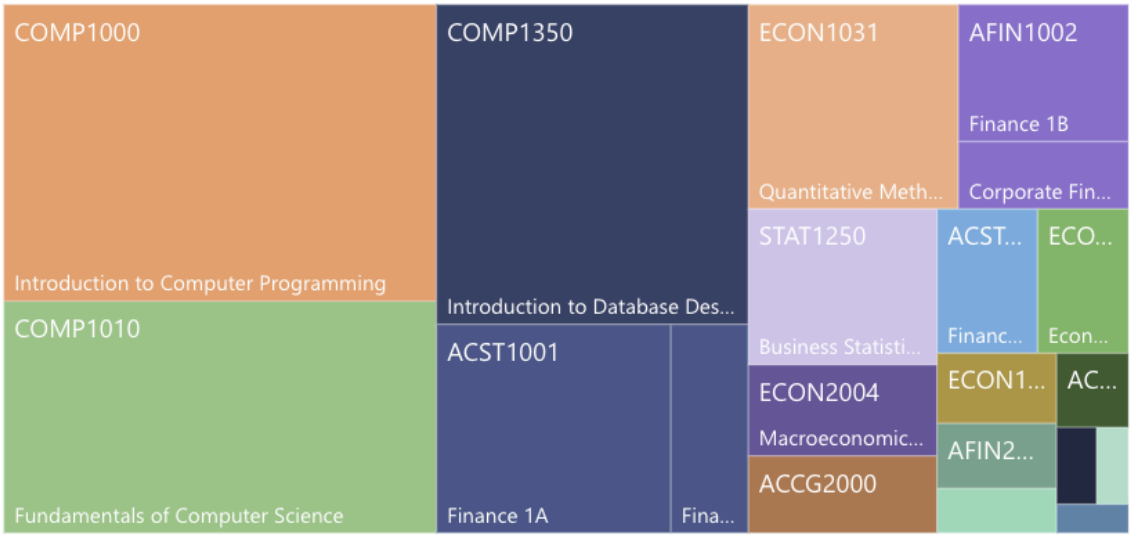
Dashboard 1: Unit Performance Trend



Repeat Fail Unit & Progress. The treemap in Figure 5 identifies units contributing most significantly to delayed course completion by trapping students in repeat-fail cycles. The findings directly mirror the analysis from Figure 2, with foundational Computer Science units COMP1000, COMP1010, and COMP1350 responsible for the highest counts of repeat-fail students. These units function as persistent structural barriers and primary drivers of delayed graduation and attrition. The concentration of repeat failures in entry-level units confirms that students who struggle initially continue to struggle, creating a compounding effect that prevents progression. Resources must be allocated to these specific units to prevent cohorts from becoming trapped in the first year.

Figure 5

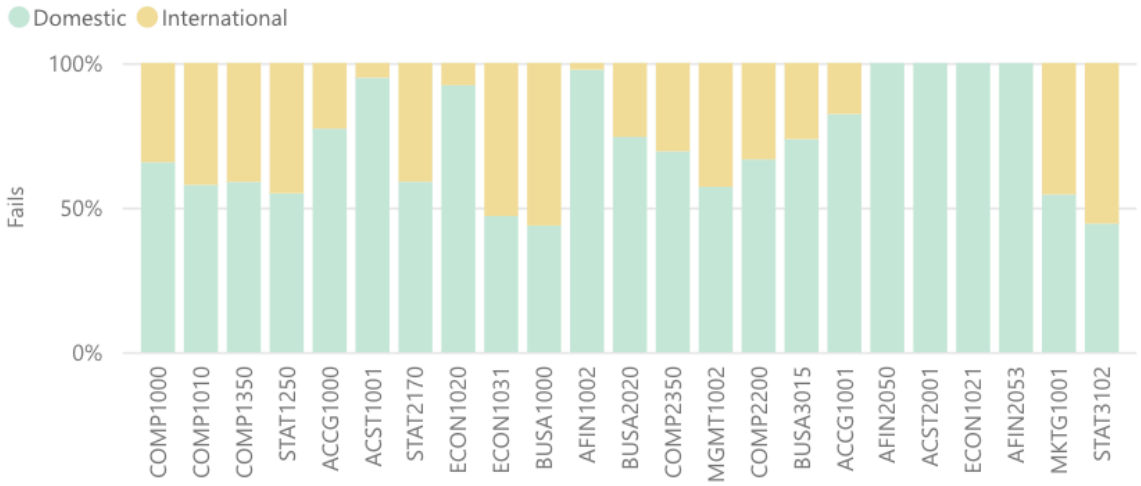
Dashboard 1: Repeat Fail Unit & Progress



Demographic Failure Split (Domestic & International). The analysis in Figure 6 reveals that structural barriers disproportionately affect domestic students. Across all units, domestic students record significantly higher failure rates than international students. This trend remains consistent in the main bottleneck unit, COMP1000, where domestic students also record higher failure rates. This suggests that foundational technical challenges present a greater barrier for domestic students, potentially due to differences in prior academic preparation. This demographic pattern indicates that intervention strategies must be tailored rather than generalised, with specific support mechanisms designed to address the distinct needs of domestic student cohorts entering foundational units.

Figure 6

Dashboard 1: Demographic Failure Split (Domestic & International)



Implications for Course Directors

The dashboard identifies common structural barriers across various technical curricula, highlighting systemic misalignment rather than isolated failures. The location and severity of these barriers differ based on curriculum design, assumed knowledge, and student demographics. The dashboard helps Course Directors determine where the primary failure point occurs, pinpoints specific units that serve as major pipeline choke points. and identifies the student cohorts most affected. This tool shifts academic practices from reactive adjustments to proactive, systemic curriculum redesign. It supports evidence-based reforms and targeted structural support, ultimately improving student progression and timely completion rates.

Dashboard 2: Student Attrition and Risk Behaviour

Research Question and Purpose

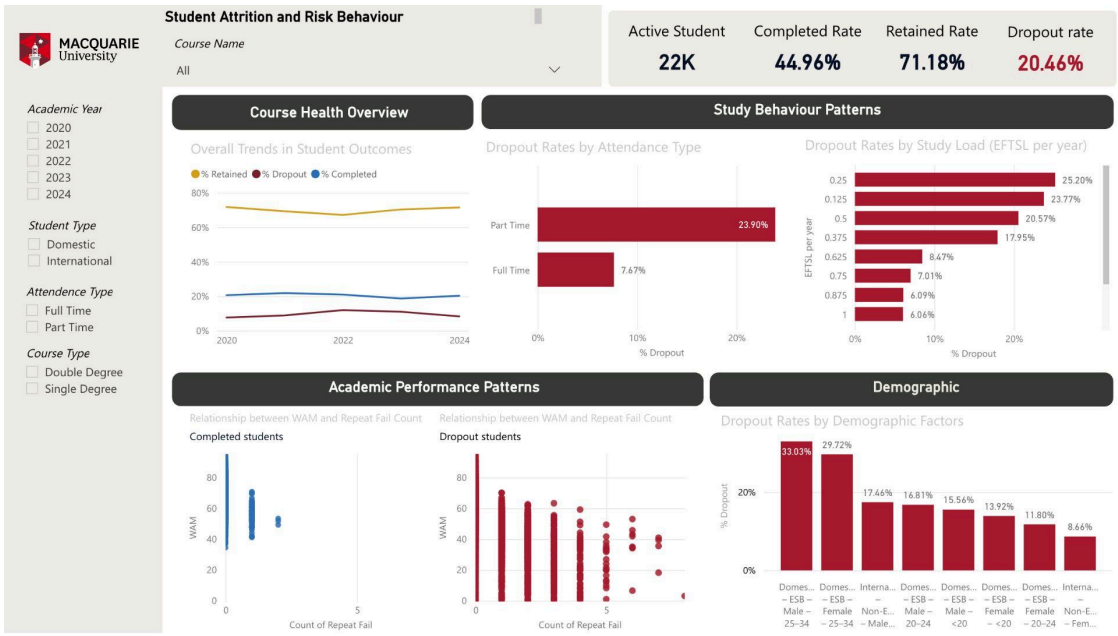
This dashboard addresses Research Question 2, which focuses on identifying early warning signals of student attrition. It aims to provide Course Directors with clear visibility into behavioural and academic patterns that lead to students dropping out. By examining attendance type, study load, academic performance, and demographic characteristics, the dashboard enables Course Directors to detect at-risk groups early and design targeted interventions that support student retention and course completion.

Dashboard Design and Layout

The dashboard is structured around four key dimensions to present a comprehensive view of student attrition, as shown in Figure 7. These dimensions capture the main drivers of dropping out, such as learning behaviour, academic progress, and individual background. The variables in each section were selected through correlation analysis, focusing on those most strongly associated with attrition. Additionally, it is designed as a dynamic and interactive tool. Filter slicers such as Course Title and Academic Year allow Course Directors to focus on courses under their responsibility, while KPI cards display quick, measurable indicators summarising key outcomes. Overall, this helps Course Directors understand underlying patterns and drivers of attrition.

Figure 7

Dashboard 2 Overview: Student Attrition and Risk Behaviour

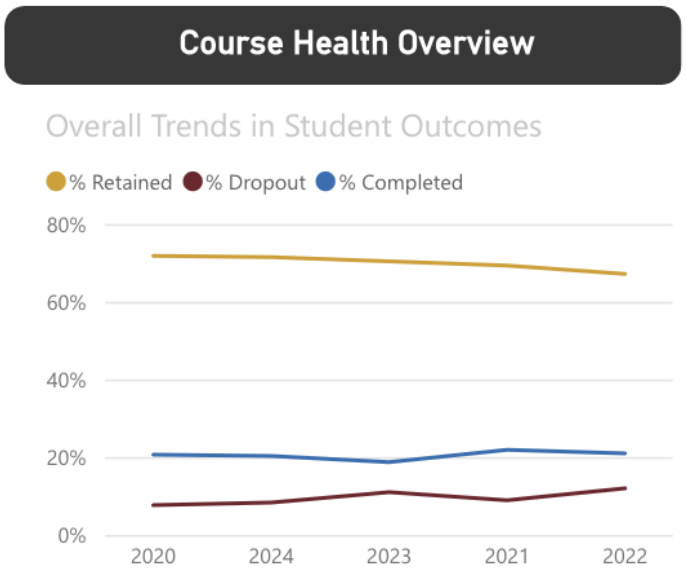


Key Findings and Interpretations

Course Health Overview. As shown in Figure 8, the line chart provides an overview of course-level performance from 2020 to 2024, displaying the proportions of students who completed, were retained, or dropped out each year to establish baseline understanding of overall student progression and stability over time. The trend lines indicate that retention remains the dominant outcome, while completion and dropout rates fluctuate moderately across the observed period. This overview serves as a contextual foundation for subsequent sections, which explore the behavioural and demographic factors contributing to these outcomes.

Figure 8

Dashboard 2: Yearly Trends in Student Outcomes (2020-2024)



Study Behaviour Patterns. Figure 9 visualises how study intensity influences attrition. The Left chart compares dropout levels between full-time and part-time students, while the Right chart examines dropout rates across different study load levels measured by year EFTSL. Attendance Type provides a broad classification of enrolment status, whereas Study Load offers a more granular view of the same behaviour.

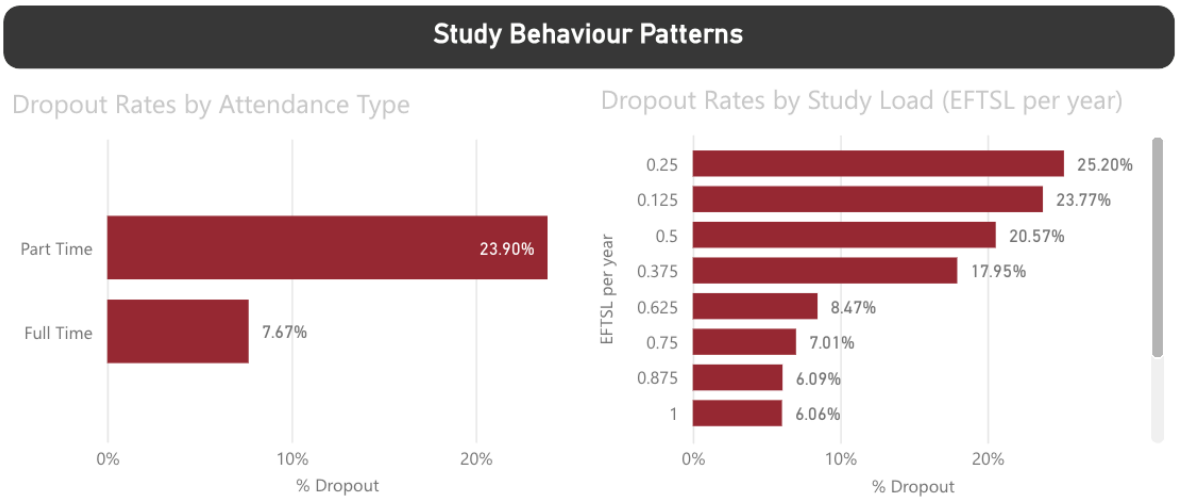
Both charts reveal a consistent pattern: Part-time students exhibit the highest risk of attrition, and this tendency becomes even clearer when examining study load, where those enrolled in low loads (below 0.5 EFTSL) show the greatest likelihood of leaving.

Collectively, these findings suggest that reduced academic engagement serves as an early behavioural signal of attrition risk.

This pattern suggests that students with lighter study loads or part-time status often face external commitments, such as work or personal responsibilities, which limit their academic engagement and may gradually reduce momentum. These insights highlight the need for targeted support for these at-risk groups and reveal a gap in data collection, as current records do not fully capture behavioural factors, such as motivation, engagement, or external pressures, that are critical for understanding and supporting this cohort more effectively.

Figure 9

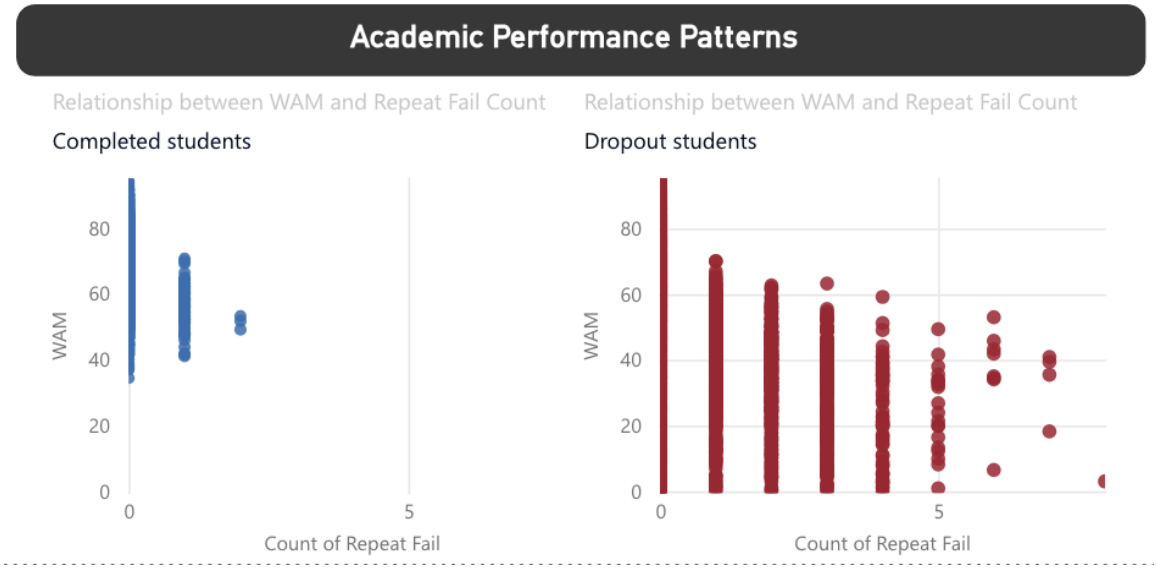
Dashboard 2 : Dropout Rates by Attendance Type and Semester Study Load



Academic Performance. The scatter plot in Figure 10 compares the academic performance profiles of completed and dropout students, clearly illustrating the contrast between successful and at-risk students. The pattern shows that students who are not retained cluster in the low-to-mid WAM range and record multiple repeated fails, whereas completed students are concentrated at higher WAM levels with minimal repetition. Notably, some students who withdrew still achieved relatively good WAM scores, suggesting that attrition is not always driven by academic ability alone but may be influenced by motivation or external factors. This finding highlights the need for targeted support for students showing early signs of disengagement, while also recognising that gaps in behavioural data limit understanding of the non-academic drivers behind attrition.

Figure 10

Dashboard 2: Relationship between WAM and Repeat Fail Count

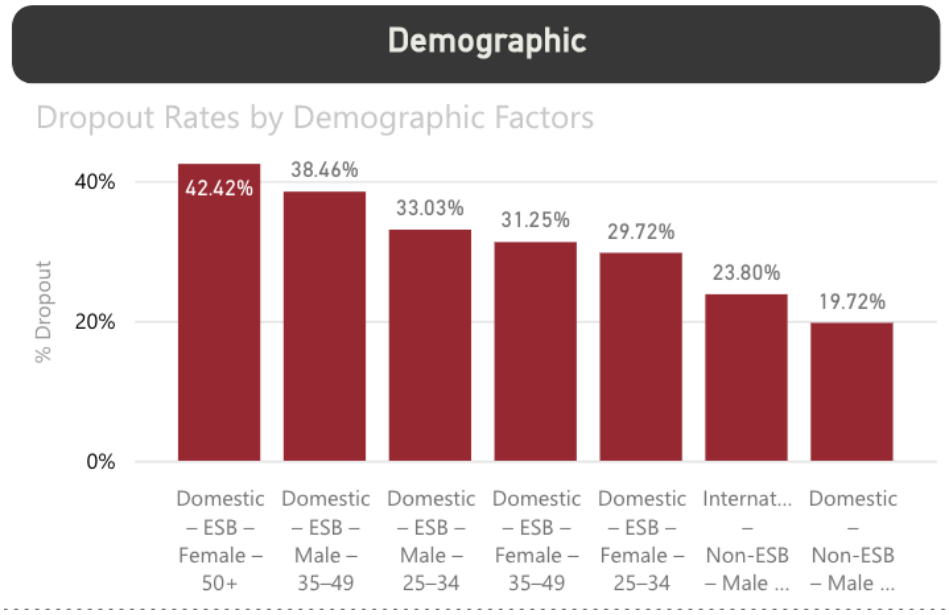


Demographic Patterns. The bar chart in Figure 11 explores how demographic attributes, including student origin, language background, gender, and age group, relate to student attrition. While specific dropout proportions may vary across courses, a consistent pattern emerges: domestic students display higher attrition rates than international students, and within this cohort, those from English-speaking backgrounds are particularly at risk.

Although these overall patterns are stable, differences in age and gender vary by course, reflecting the diverse nature of student cohorts. For instance, older students may face competing demands such as employment or family responsibilities, while gender differences may be shaped by discipline-specific learning environments. The dashboard enables each Course Director to identify the unique demographic characteristics of at-risk students within their own programme, investigate underlying causes, and implement timely, targeted support measures to improve retention outcomes.

Figure 11

Dashboard 2: Dropout Rates by by Demographic Factors



Implications for Course Directors

The dashboard reveals that attrition is driven by interconnected factors rather than a single cause. Lower study intensity, repeated academic failure, and specific demographic vulnerabilities interact to heighten dropout risk. This integrated view helps Course Directors understand how learning behaviour, performance, and background combine to shape student outcomes, turning the dashboard from a descriptive tool into a foundation for early risk detection.

Patterns observed in the dashboard are largely consistent across courses, reflecting learning behaviours rather than discipline-specific differences. However, the relative impact of these behaviours can vary depending on curriculum design, workload intensity, and assessment structures. In most cases, what differs is the degree of these patterns rather than their nature.

Building on these insights, the dashboard provides a clear baseline for comparison and benchmarking. Course Directors can examine whether their course's dropout rate, study load, or academic performance levels sit above or below the faculty average and identify which risk factors are most pronounced within their own cohort. By doing so, the tool supports proactive identification and early intervention, enabling data-informed decisions and continuous improvement in student progression and retention outcomes.

Predictive Analytics and Modeling

The predictive model was developed to extend the descriptive analysis in the dashboards to predictive analytics. Using historical data from undergraduate students enrolled in Macquarie Business School courses between 2020 and 2025, the model identifies students in the 2025 cohort at risk of dropping out in 2026. By predicting at-risk students early, the model shifts the action from understanding risk to anticipating and preventing it.

Model Selection Rationale

This project used logistic regression for prediction because it is well-suited for binary classification problems: Dropout (1) and Retained (0). Logistic regression provides interpretable coefficients that indicate how each feature influences dropout probability. In addition, it facilitates communication with non-technical university stakeholders, which is essential for practical decision-making.

Data Preparation

A clean dataframe called student_features_model was created, including each student's most recent study record. A function was used to decide every student's status. If a student in 2025 had the code "Not Retained in Year X+1", the model would look back at their 2024 record to obtain the actual result, as 2026 enrolment data is not yet available. Students without prior records were marked as "NaN" and excluded from the model.

For prediction, another dataset was made using students still studying in 2025 whose status was not yet known. These students were given a predicted dropout risk score by the trained model to identify who may need extra support in the following year.



Features Used in the Model

The features used in model training came from the student performance dataset (student_features_model), focusing on the most relevant predictors based on correlation with the dropout variable. After feature ranking and correlation analysis, the top five features showing the strongest relationship with student dropout were selected.

- WAM_SCALED: measures academic performance
- PROGRESS_SCALED: percentage of course completion
- FAIL_COUNT_SCALED
- REPEAT_FAIL_COUNT_SCALED
- WITHDRAWAL_COUNT_SCALED

Feature coefficients showed that Progress (−0.58) and WAM (−0.52) had the strongest negative correlations with dropout, which means the students with low progress or poor grades are more likely to drop out. On the other hand, Fail Count (+0.33) and Repeat Fail Count (+0.23) increased dropout probability.

Model Training and Validation Approach

The data was split into 70% training and 30% testing subsets. The logistic regression model was trained using the scaled features and the target variable DROPOUT. Model performance was evaluated using multiple metrics. The model reached about 86%

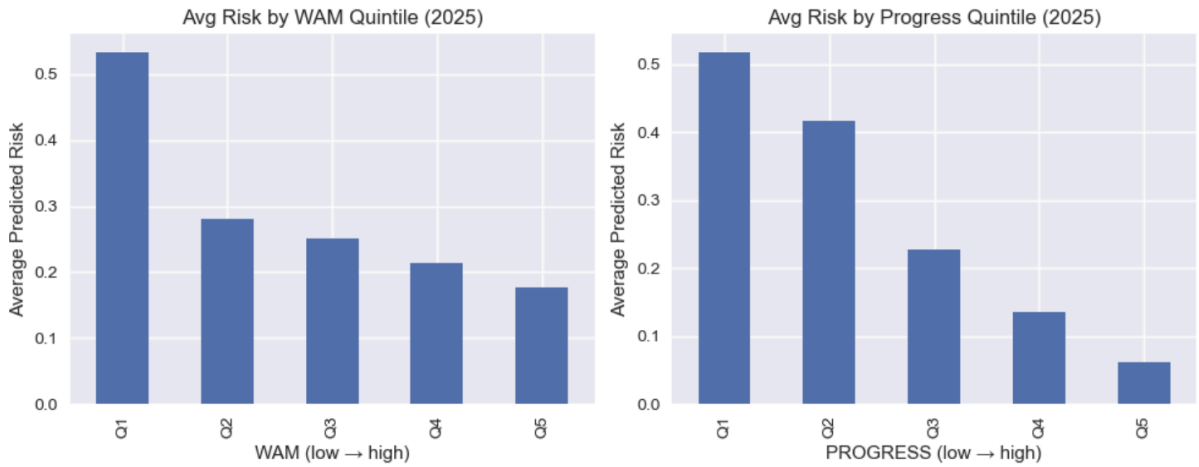
accuracy, showing that most predictions were correct. The AUC score is 0.90, which can confirm that the model could successfully separate high-risk from low-risk students.

Insights from the model

The model identified 285 (about 4%) of the 7,079 active students in 2025 as high-risk for the following year, with dropout probability greater than 80%. Visualisation of average predicted risk distribution (see Figure 12) shows that lower grades and lower progress lead to higher dropout probabilities.

Figure 12

Average Predicted Dropout Risk by WAM and Progress Quintiles



Insights Summary and Recommendations

This section synthesises findings from diagnostic dashboards and predictive modelling to present three key insights and evidence-based recommendations for course directors and institutional stakeholders.

Insights Summary

1. Foundation Units Require Comprehensive Review

First-year foundation units exhibit the highest failure and repeated-failure rates across the program. While quantitative analysis clearly identifies performance issues within foundation unit content or delivery, qualitative factors likely contribute to these patterns. First-year students require substantial support during the transition from secondary to tertiary education, where academic expectations and lifestyle demands differ significantly. Students must develop greater independence and organisational capacity. This transition challenge represents a behavioural dimension not captured in available datasets.

2. Four Key Features Serve as Primary Risk Indicators

Integrated analysis from Dashboard 2 and predictive modelling identifies four compounding risk factors that substantially elevate dropout probability: low Weighted Average Mark, reduced study load, minimal progress percentage, and patterns of repeated unit failures.

3. Domestic Students Require Targeted Support

Domestic students from English-speaking backgrounds experience disproportionate challenges, particularly within foundation technical units, indicating systematic barriers requiring targeted intervention.

Recommendations

1. Review Foundational Units and Redesign Accordingly

Course directors should conduct granular analysis of individual foundation units to identify specific drivers of student disengagement. Each unit may require distinct interventions, including content revision, workload recalibration, assessment restructuring, unit coordinator changes, or seminar class size reduction.

2. Enhance First-Year Transition Support

Macquarie Business School should establish a peer mentoring program connecting first-year students with trained mentors from years 2-4. At the institution level, the O-Week programming should emphasise practical guidance on university navigation, academic expectations, and support service access. While comprehensive support infrastructure exists, student awareness and utilisation remain critical success factors.

3. Implement Tiered Intervention

Course directors should deploy the predictive model to stratify students into three risk categories (high, medium, low) and implement differentiated intervention protocols. High-risk students require intensive support including mandatory academic advising, career counselling, and mental health service referrals where appropriate. Medium-risk students benefit from structured peer support networks and regular progress monitoring. Low-risk students receive preventive resources and proactive engagement communications. Critically, intervention strategies must address all risk tiers to prevent escalation from low to medium and medium to high-risk categories.



Conclusion

This project aimed to help Course Directors better understand student progression and retention challenges amongst Bachelor students at Macquarie Business School. By analysing enrolment and retention data, building dashboards, and training a predictive model, this work provides clear insight into where students struggle and who is most at risk of leaving. The dashboards highlighted early curriculum barriers, particularly in first-year core units, and revealed that low study load, repeated failures, and certain demographic factors strongly correlate with attrition. The logistic regression model further strengthened these insights by predicting high-risk students with strong accuracy, equipping Course Directors with the capability to intervene before dropout occurs.

This work delivers practical value to the Business Intelligence and Reporting team by supporting data-informed decisions and early intervention strategies. However, several limitations exist. Missing fields in the dataset, gaps in behavioural data, and the application of only one predictive model constrain the scope of this analysis. Future work would benefit from collecting comprehensive behavioural data, expanding model features, applying alternative machine learning techniques, and integrating real-time student engagement data to improve predictive accuracy. Overall, this project supports student success and strengthens institutional planning and support services.

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Appendix A

Dashboard and Predictive Modelling Files

Dashboard 1 Curriculum Barrier Analysis.pbix:

https://drive.google.com/file/d/1kWMVAs9YlGwuBONh76v2CubwBuorALJW/view?usp=sharing

Dashboard 2 Student Attrition and Risk Behaviour.pbix:

https://drive.google.com/file/d/1laVz0gejqleTe-J9jsEeZ8nf7wpe4BDb/view?usp=sharing

Data Preparation and Predictive Modelling.ipynb

https://drive.google.com/file/d/19j09YqcuhI9BAicejGr0x3kH4e37eQCZ/view?usp=sharing

Appendix B

Acknowledgment Statement of Use of GAITs

We acknowledge that we have only used GAITs (e.g., ChatGPT, Google Gemini, Claude) in drafting, proofreading, and clarifying this assignment, which is permitted in the assignment instructions.